

CONEPRO · DESIGN PRECIS

ConePro design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/conepro		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.8	2026-10-02	Amish Chadha	Decisions of 2026-10-02 carried in: end cap for the upper rod, T-handle grips, hammer grip grooves, first co-design partner to approach
0.7	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.6	2026-09-30	Amish Chadha	Design for construction (CNP-DDR-004, open for Amish's review); M12 rod joints, one-piece clamp against the anvil, reel internals and fixings, reel-to-logger cable (BOM line 17); stroke 939 mm; \$335; numbers from CNP-CAL-001 v0.4; CNP-DWG-001 Rev P5; build plan CNP-BLD-001
0.5	2026-09-27	Amish Chadha	Reel housing 72 x 60 x 72 mm round the 60 mm drum (CNP-DDR-003, decided by Amish on 2026-09-27); stroke 928 mm, 850 mm usable; numbers from CNP-CAL-001 v0.3; CNP-DWG-001 Rev P3
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002); 6 mm plate, aluminum

			clamp and lighter bag (15.7 kg), optional extension rod, reel details accepted, stiff-ground guidance; numbers from CNP-CAL-001 v0.2; CNP-DWG-001 Rev P2
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices decided (CNP-DDR-001); numbers replaced by CNP-CAL-001 results; parametric model, drawing CNP-DWG-001 and refreshed media
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

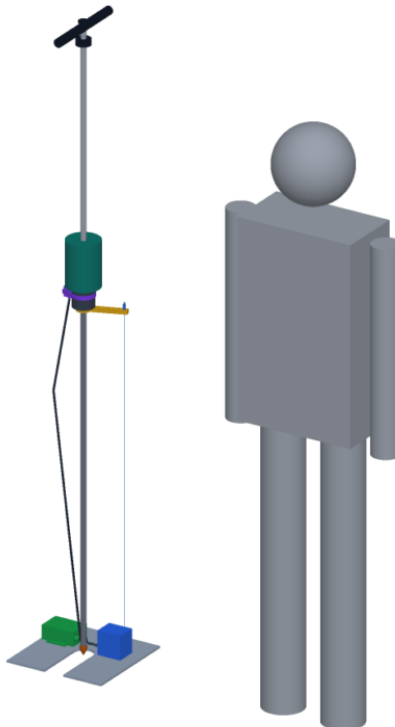
ConePro design precis

ConePro is a standard-geometry dynamic cone penetrometer (8 kg hammer, 575 mm drop, 20 mm 60 degree cone, as in ASTM D6951) with a draw-wire depth sensor referenced to a plate on the ground, a clamp-on sensor pad at the anvil that detects blows and reads rod tilt, and a small BLE logger, so one person can run a test while the phone plots penetration against blows and exports the record. The sensor set is designed to fit existing standard penetrometers too. The TRL 3 calculations (CNP-CAL-001 v0.4) give about 28 J at the cone per blow, 850 mm of usable penetration per lower rod, about 40 h of logging on three AA cells and \$335 in parts, \$65 under the \$400 value-engineering target (a hypothetical control target, not a limit). With the changes Amish accepted on 2026-09-25 (CNP-DDR-002), the instrument carried in its bag weighs about 15.7 kg (was 16.7 kg) and meets the 16 kg target (R10), and a 500 mm extension rod is a separately carried accessory for tests deeper than 850 mm (R5). On 2026-09-27 Amish decided to keep the 60 mm drum and grow the reel housing to 72 x 60 x 72 mm (CNP-DDR-003), which trimmed the stroke from 940 mm to 928 mm. On 2026-09-30, under Amish's instruction to make the design physically buildable, the model was made constructable (CNP-DDR-004, accepted by Amish on 2026-10-02 with one exception: no thread locker on the rod, anvil and cone joints): threaded rod joints, a one-piece clamp that butts the anvil, a reel with its drum, spring motor and sensor inside, fixings for every part and a cable from the reel to the logger. The stroke is now 939 mm, and the prototype build plan (CNP-BLD-001, [docs/05-build-plan.md](#)) shows how to make and assemble every part. No requirement is now not met; depth accuracy (R2) and shock survival (R11) remain at risk.

ConePro

github.com/BoujeeEnjinia1701/conepro

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey figure: 1.75 m person for scale

Figure 1. ConePro set up at the start of a test, with a 1.75 m person for scale. The cone tip rests on the ground through the slotted reference plate and the hammer rests on the anvil.

■ How it works

1. **Set up.** The operator lays the slotted reference plate on the ground, stands the cone in the plate's center hole, clamps the draw-wire arm to the rod just below the anvil and straps the sensor pad to the anvil. The app zeroes the depth with the cone point at the surface, and zeroes the tilt by turning the rod half a turn.
2. **Drive.** The operator holds the handle, lifts the 8 kg hammer to the top stop and lets it fall 575 mm onto the anvil. Each blow drives the rod and cone further into the soil.
3. **Sense.** A ring of magnets in the hammer face passes a potted Hall-effect switch in the sensor pad as the hammer lands, and the pad's accelerometer sees the impact peak. A blow counts only when both fire and the depth steps down or holds within 0.5 s. Between blows the accelerometer reads rod tilt. A draw-wire sensor on the plate measures how far the clamp arm has moved down, which equals the cone's penetration because the rod is rigid and the plate stays on the ground surface.
4. **Log.** The logger on the plate reads depth at 100 Hz, waits 0.2 s for the rod and wire to settle after each blow, and stores blow number, depth, tilt and time. It keeps the record even when no phone is connected.
5. **Show.** The phone app plots depth against blows, corrects depth for rod lean, splits the curve into layers of near-constant slope, and gives each layer's DCP index (mm per blow) and an indicative CBR from the ASTM D6951 correlations, with the correlation named on screen. It warns when the rod leans more than 5 degrees. The user exports a CSV with time, location and notes.
6. **Extract.** After the test the operator removes the plate over its slot and pulls the rod out with the optional lever rod puller, which is carried separately. For tests deeper than 850 mm the operator adds the optional 500 mm extension rod and the app re-zeroes. In stiff ground where the lever will not free the rod, the user guidance follows the disposable-cone practice: drive a disposable cone that stays in the ground, so the rod comes out through a hole slightly larger than itself (CNP-DDR-002, D12).

ConePro: energy per blow, J (estimates)

github.com/BoujeeEnjinia1701/conepro

CONCEPT, NOT FOR FABRICATION

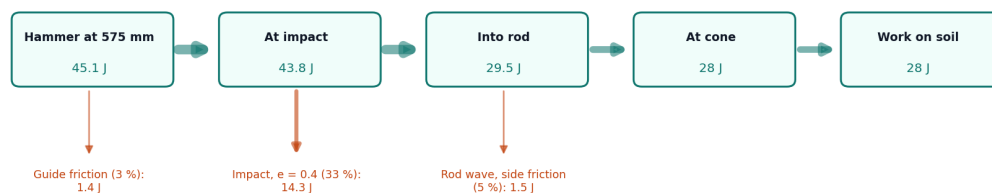


Figure 2. Energy per blow from hammer to soil, in joules (estimates from CNP-CAL-001, section B). The impact loss depends on the restitution between hammer and anvil, which is assumed, not measured.

■ Main components

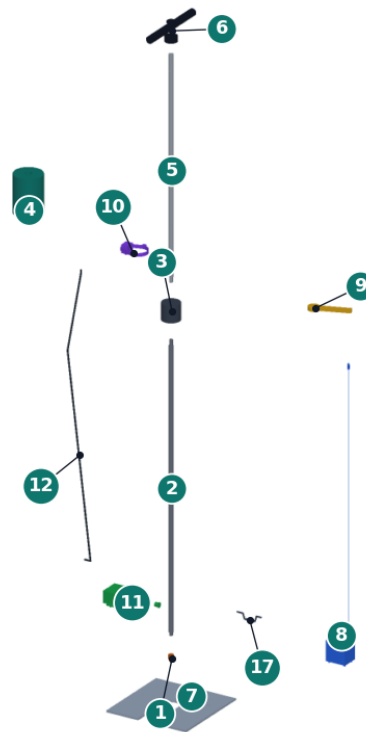
Numbers match the exploded view (Figure 3) and `bom/bom.csv`. Geometry is in the parametric model `cad/src/model.py` and on drawing CNP-DWG-001.

#	Component	Choice	Notes
1	Cone	Hardened steel, 20 mm base, 60 degree point, 12 mm shoulder tapped M12, screwed onto the rod	Replaceable; a worn cone biases results
2	Lower drive rod	16 mm steel, 1,000 mm between shoulders, M12 studs on both ends, engraved every 10 mm	Engraving is a manual backup scale and the calibration reference
3	Anvil and coupler	Steel, 64 mm diameter x 60 mm, tapped M12 at both ends	Plain; the sensor pad clamps on (D3)
4	Drop hammer	8.0 kg steel, 100 mm OD x 22 mm bore x 136.5 mm	Sixteen 8 x 4 mm magnets bonded in pockets in the lower face, outside the strike area; grip grooves, with the length set at machining to keep 8.00 kg (decided 2026-10-02)
5	Upper rod (hammer guide)	16 mm steel, 776 mm above the anvil plus an M12 stud, running up through the handle	Gives the 575 mm free drop to the stop
6	Handle and top stop	26 mm tube T-handle welded to the upper rod, 44 mm stop collar on a 6 mm roll pin, bubble level on a welded seat	Level is the tilt backup (D1); rubber grips about 33 mm across on the T-handle (decided 2026-10-02)
7	Reference plate	300 x 300 x 6 mm aluminum, 60 mm center hole and slot	Depth reference on the ground surface; 6 mm to save mass (D9)
8	Draw-wire depth sensor	60 mm grooved aluminum drum, single layer, constant-force spring, 12-bit angle sensor, groove keeper, preloaded eye spring	Drum must be metal for thermal stability; bought industrial sensor is the fallback
9	Anvil clamp and wire arm	One piece cut from 12 mm aluminum plate: split collar on the lower rod, butting the anvil underside, arm about 115 mm to the wire eye	Wire exit on the plate 105 mm from the rod; aluminum to save mass (D9)
10	Blow and tilt sensor pad	Hall switch and 3-axis accelerometer, potted, on a 300 Hz elastomer isolator, band clamp for 50 to 80 mm anvils	Top 8 mm below the anvil top, under the hammer overhang
11	BLE logger	Microcontroller with BLE, 4 MB flash, button and LED, three AA cells, IP65 box	Kept on the plate, away from impacts
12	Coiled sensor cable	Pad to logger, with a strain relief at each end	Follows the rod down during the test
17	Reel-to-logger cable	4-core cable through glands, in two P-clips on the plate	Carries the angle sensor signal; added for construction (CNP-DDR-004)
15	Rod extraction lever (optional)	Lever puller with a self-gripping 16 mm rod clamp, about 10:1, about 3 kg	Carried separately; outside the R13 total (D2)
16	Extension rod (optional)	16 mm steel, 500 mm, M12 studs and a 20 mm coupling sleeve, engraved every 10 mm, 0.79 kg	Carried separately with the lever; outside the R10 mass and R13 total (D10)

ConePro: exploded viewgithub.com/BoujeeEnjinia1701/conepro

CONCEPT, NOT FOR FABRICATION

- 1 Hardened cone, 20 mm, 60 degree
- 2 Lower drive rod, 16 mm x 1,000 mm
- 3 Anvil and coupler
- 4 Drop hammer, 8 kg
- 5 Upper rod (hammer guide)
- 6 Handle, top stop, bubble level
- 7 Reference plate, slotted
- 8 Draw-wire depth sensor
- 9 Anvil clamp and wire arm, aluminum
- 10 Blow and tilt sensor pad
- 11 BLE logger, 3 x AA
- 12 Coiled sensor cable
- 17 Reel-to-logger cable



Seen from the front right and above, 24 deg elevation; numbers match bom/bom.csv

Figure 3. Exploded view with numbered callouts matching the BOM. The carry bag, hardware and the optional extraction lever and extension rod are not shown.

■ Key numbers

All values are estimates from CNP-CAL-001; the bracketed tag names the line of `docs/04-calcs/sizing.py` that prints each one.

Table 1. Blow energy and timing.

Quantity	Estimate	Basis
Potential energy per blow	45.1 J [B1]	8.0 kg x 9.81 m/s ² x 0.575 m
Impact speed; fall time	3.36 m/s; 0.342 s [B1]	Free fall over 575 mm
Energy reaching the cone	28.0 J, 62 % (26.1 to 31.2 J) [B4]	3 % guide friction; Hiley impact efficiency with restitution 0.4 (0.2 to 0.6) and 5.09 kg driven mass; 5 % rod and side friction
Mean dynamic soil resistance	1.9 kN at 15 mm per blow; 14.0 kN at 2 mm per blow [B6]	Energy at the cone over penetration
Blow rate	17 to 24 per minute	One blow every 2.5 to 3.5 s (assumed)

What the numbers mean for the user. The ASTM D6951 correlation for most soils is $CBR = 292 / DCP^{1.12}$, with DCP in mm per blow (separate correlations apply to lean clays below CBR 10 and to fat clays). Table 2 shows how sharply the indicative CBR changes at small penetrations, which is why depth accuracy matters most in stiff layers.

Table 2. Indicative CBR from the ASTM D6951 general correlation [E7].

DCP index (mm per blow)	2	5	10	15	25	50
Indicative CBR (%)	134	48	22	14	8	4

Table 3. Depth measurement.

Quantity	Estimate	Requirement
Resolution	0.046 mm per count [E1]	R2 resolution met
Bench error over 1,000 mm, vertical rod	±0.98 mm worst case; ±0.49 mm root sum square [E3]	R2 (±1 mm) at risk
Extra error from rod lean at 850 mm, after tilt correction	0.95 mm at 1 degree; 1.79 mm at 2 degrees [E4]	Why the usable range stops at 850 mm (accepted, D11)
DCP index error, 10 blows at 2 mm per blow	±4.9 % RSS (±9.8 % worst); CBR ±5.5 % [E8]	
Wire slack after a blow	1.9 to 17 mm for 5 to 35 ms [E5]	Groove keeper and preloaded eye spring needed
Record latency	0.21 s after impact [E6]	R4 met
Plate settlement or heave	Not calculable; static pressure only 0.24 kPa [E9]	Open question

Table 4. Range, mass, time, shock and power.

Quantity	Estimate	Requirement
Stroke per lower rod	939 mm, limited by the wire arm meeting the reel housing (72 x 60 x 72 mm round the 60 mm drum, CNP-DDR-003); 850 mm usable [D1, D2]; optional 500 mm extension rod [C7]	R5 (restated, D10) met
Assembled height	1,874 mm [A3]	
Mass as carried in the bag	15.7 kg (was 16.7 kg); 15.3 kg without the bag [C2, C6]	R10 (16 kg) met, thin
Heaviest piece	9.83 kg, hammer captive on the upper rod [C3]	R10 (10 kg) met
Packed length	About 1.08 m (longest piece 1,029 mm) [C4]	R10 met
Reference test time	8.1 to 11.7 min [J1]	R9 met, thin
Shock at the anvil; at the isolated pad	1,400 to 5,500 g mean; about 544 g [G1, G2]	R11 at risk
Tilt reading	±0.3 degree after rotation zeroing [H1]	R6 met by design
Average current; battery life	50.2 mA; 40 h, 20 h at 0 °C [K1, K2]	R12 met
Storage	About 830 tests in 4 MB [I1]	R8 met

Table 5. Cost [L1].

Group	Indicative cost
Mechanical DCP (items 1 to 7)	\$174
Sensing and logging (items 8 to 12, 17)	\$113
Carry bag, hardware and consumables (items 13, 14)	\$48
Instrument total	\$335 (R13, within the \$400 value-engineering target)
Optional extraction lever (item 15) and extension rod (item 16)	\$50 and \$16, outside the instrument total

■ Key design choices

Amish decided the TRL 2 review items on 2026-09-25 by accepting every recommendation (CNP-DDR-001). The choices below are therefore decided, not proposed.

- **ASTM D6951 geometry (D5).** The standard 8 kg hammer and 575 mm drop, so users can apply published correlations and compare with existing DCP data. The 4.6 kg option is noted for later.
- **Depth by draw-wire from a ground plate (D4).** A draw-wire reel on the plate measures the clamp arm's movement. A laser time-of-flight sensor is the fallback to compare later. TRL 3 adds three details to meet the calculations: a metal, single-layer grooved drum; a groove keeper for the slack after each blow; and a preloaded spring at the arm eye to limit the snap tension.

- **Blow detection by Hall sensor with cross-checks (D6).** A blow counts when the Hall switch sees the hammer arrive, the accelerometer sees an impact peak and the depth steps down or holds within 0.5 s. A peak below half the full-drop value flags a short drop.
- **Tilt by accelerometer, with a bubble level as backup (D1).** The accelerometer in the sensor pad reads tilt between blows, and the app warns above 5 degrees and corrects depth for lean.
- **Electronics on the plate (D7).** Only the isolated sensor pad sees impacts.
- **Three AA cells (D8).** No lithium safety or shipping issues; sold everywhere.
- **Optional lever for extraction (D2).** A lever puller with a rod clamp, carried separately and outside the instrument cost and mass.
- **Sensor set fits standard DCPs (D3).** The clamp, band-mounted pad, plate and reel need no machining of the penetrometer (R15). The prototype is a complete instrument.

The TRL 3 calculations raised four further items, which Amish decided on 2026-09-25 by accepting every recommendation (CNP-DDR-002):

- **Lighter carried set (D9).** A 6 mm aluminum plate (1.29 kg, was 1.72 kg), an aluminum clamp and arm (0.12 kg, was 0.34 kg) and a 0.40 kg nylon roll bag (was 0.80 kg) bring the carried mass from 16.7 kg to 15.7 kg, within R10.
- **Extension rod as an accessory (D10).** R5 is restated as 850 mm per lower rod. A 500 mm extension rod is carried separately with the lever for deeper tests.
- **Reel details accepted (D11).** The metal single-layer grooved drum, groove keeper and preloaded eye spring stay in BOM line 8, and the usable range stays at 850 mm per rod. The bought industrial draw-wire sensor remains the fallback.
- **Stiff ground (D12).** No hardware change; the user guidance states the disposable-cone practice for ground where the lever cannot free the rod.

■ Safety

Safety: ConePro drops an 8 kg steel hammer onto a steel anvil every few seconds and drives a pointed steel rod into the ground. Treat buried services, crushed fingers, flying fragments, noise, back strain and misuse of the results as hazards at every test point.

- **Buried services.** A rod driven 1 m into the ground can strike electric cables, gas pipes and water mains. Get a utility locate before every test (in the United States, call 811 or use the state one-call service) and do not test within the marked tolerance zone.
- **Crushed fingers and hands.** The hammer lands with about 44 J. Keep hands on the handle only, never on the anvil, the hammer's lower face, the sensor pad or the clamp arm. The pad sits 8 mm under the hammer overhang, which is a pinch point: fit and remove it only with the hammer held at the top stop or removed.
- **Flying fragments and mushrooming.** Repeated steel-on-steel impact at thousands of g can chip the anvil or hammer face and spall hardened steel. Wear safety glasses, inspect the faces, and replace them when they mushroom or crack.
- **Magnets.** The hammer carries sixteen strong magnets. Keep it away from pacemakers and other implanted devices, and from cards and instruments that magnets can damage.

- **Draw-wire.** A wire that breaks or is released under tension can whip. Keep faces away from the reel and never let the arm snap back.
- **Noise.** Steel-on-steel impact is loud at close range. Hearing protection is recommended; the level is not yet estimated.
- **Lifting and posture.** The hammer is lifted about 57 times per test and the upper assembly weighs about 9.8 kg. Use a straight back and swap operators on long test days. Pull the rod only with the lever, never by back strength; stop if the lever will not move it (the pull force can exceed 3 kN in stiff ground), and use a disposable cone for further tests at that site (D12).
- **Rod whip and tipping.** A leaning rod can bend or kick sideways when struck. Stop when the app warns of lean (R6).
- **Hammer sliding off when packed.** Whenever the upper rod is off the anvil, a screw-on end cap on its M12 stud stops the 8 kg hammer sliding off onto the user's feet; a label on the hammer says so (decided 2026-10-02).
- **Sharp cone.** Cover the cone in transport.
- **Ground and site hazards.** Contaminated land, trench edges and traffic: follow site rules and do not test at the edge of an open excavation.
- **Misuse of results.** CBR values are indicative correlations. ConePro results must not be used on their own to design foundations or to decide that ground is safe to build on; they show where a qualified geotechnical engineer should investigate.
- **Batteries.** Alkaline AA cells carry little risk; remove them for storage to avoid leakage.

■ Open questions

- Approach the first co-design partner: a low-volume road agency (a county or district road department) that already runs manual DCP tests on unpaved roads, with a university pavement or geotechnical lab as the second choice (decided 2026-10-02; not yet agreed).
- Measure the restitution, the magnet field with the steel hammer, the isolator response and the draw-wire error on a bench.
- Estimate plate settlement or heave under blows, and how to detect it (for example a second reference point).
- Define the blow and depth data format and the layer-splitting method; check against ORN 8 practice.
- Confirm commercial prices for instrumented penetrometers (not yet checked online).

Concept media: [blueprint sheet](#), [interactive 3D model](#). General arrangement: [CNP-DWG-001](#).